pipeline, generating a response directly; it returns the response to the previous middleware, and the endpoint middleware never sees the request. This scenario is an example of the *chain-of-responsibility* design pattern.

DEFINITION When a middleware component short-circuits the pipeline and returns a response, it's called *terminal middleware*.

A key point to glean from this example is that the pipeline is *bidirectional*. The request passes through the pipeline in one direction until a piece of middleware generates a response, at which point the response passes *back* through the pipeline, passing through each piece of middleware a *second* time, in reverse order, until it gets back to the first piece of middleware. Finally, the first/last piece of middleware passes the response back to the ASP.NET Core web server.

The `HttpContext` object

I mentioned the `HttpContext` in chapter 3, and it's sitting behind the scenes here, too. The ASP.NET Core web server constructs an `HttpContext` for each request, which the ASP.NET Core application uses as a sort of storage box for a single request. Anything that's specific to this particular request and the subsequent response can be associated with and stored in it. Examples are properties of the request, request-specific services, data that's been loaded, or errors that have occurred. The web server fills the initial `HttpContext` with details of the original HTTP request and other configuration details, and then passes it on to the middleware pipeline and the rest of the application.

All middleware has access to the `HttpContext` for a request. It can use this object to determine whether the request contains any user credentials, to identify which page the request is attempting to access, and to fetch any posted data, for example. Then it can use these details to determine how to handle the request.

When the application finishes processing the request, it updates the `HttpContext` with an appropriate response and returns it through the middleware pipeline to the web server. Then the